Airport Investment Intelligence Agent

Timeframe: 24 hours

Goal

We are a firm that invests in airport modernization projects in the US. We are exploring how AI agents can help analysts identify promising airport investment opportunities. The goal of the agent is to help us identify airports where renovations will be most profitable based on increased flight and passenger capacity.

Instructions

Build an AI-powered agent that can answer questions such as:

- Which airports in New England are strong candidates for terminal expansion?
- Compare LA and Santa Ana airport congestion levels.
- What is the percentage of long haul flights out of Anchorage airport?
- What is the unmet flight demand in SFO airport and why?

The Agent Should:

- Use public APIs to gather airport/aviation data
- Rank or compare airports based on your defined logic or KPI
- Explain its reasoning clearly
- Support conversational follow-up questions

Requirements:

- Include some deterministic scoring or ranking logic (not only LLM output)
- Include a chat interface to talk with the agent (voice is a bonus)
- Clearly communicate assumption, uncertainty and scoping

Deliverables:

- Source code
- Short design/architecture document explaining:
    - Scoring methodology
    - Key tradeoffs
    - Where/how AI is used

This assignment is intended to be achievable in approximately one day. Prioritize clarity, reasoning, and thoughtful design over completeness or polish.